INCH-POUND
MIL-DTL-18110G(SH)
5 March 2018
SUPERSEDING
MIL-V-18110F(SH)
19 February 1991

DETAIL SPECIFICATION

VALVES, GATE TYPE, STEEL
SIZES 2½ INCHES TO 16 INCHES)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steel gate type valves, in sizes from 2½ to 16 inches, for shipboard steam, water, and oil services.

1.2 Classification. Valves are of the following compositions, ratings and classes, and sizes, as specified (see 6.2).

1.2.1 Compositions. Valves are of the following compositions (see table I for full details):

a. Composition A:
   (1) Chromium – 2¼ percent.
   (2) Molybdenum – 1 percent.

b. Composition B:
   (1) Chromium – 1¼ percent.
   (2) Molybdenum – ½ percent.

c. Composition D:
   (1) Carbon steel.

d. Composition E:
   (1) Corrosion-resistant steel.

1.2.2 Ratings and classes. Unless otherwise specified herein, valves are constructed and rated in accordance with ASME B16.34, as follows:

a. Standard rating:
   (1) Class 150
   (2) Class 300
   (3) Class 400

b. Special rating:
   (1) Class 600

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(2) Class 900

(3) Class 1,500

(4) Class 2,500

Composition E valves are of standard ratings only, classes 150, 300, and 400.

1.2.3 Sizes. Valves are of the following nominal pipe sizes (nps):

a. 2½ inches
b. 3 inches
c. 4 inches
d. 5 inches
e. 6 inches
f. 8 inches
g. 10 inches
h. 12 inches
i. 14 inches
j. 16 inches

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-1222	-	Studs, Bolts, Screws and Nuts for Applications Where a High Degree of Reliability is Required; General Specification for
MIL-DTL-15024	-	Plates, Tags, and Bands for Identification of Equipment, General Specification for
MIL-P-15024/5	-	Plate, Identification
MIL-R-17131	-	Rods and Powders, Welding, Surfacing
MIL-P-24503	-	Packing Material, Graphitic, Corrugated Ribbon or Textured Tape and Preformed Ring
MIL-P-24583	-	Packing Material, Graphitic or Carbon Braided Yarn
DOD-V-24657	-	Valve Actuator, Direct Coupled, Gear Driven Electrically Powered for Naval Shipboard Propulsion and Auxiliary Systems (Metric)
MIL-DTL-24696	-	Gasket, Sheet, Non-Asbestos

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MIL-G-24716 - Gaskets, Metallic-Flexible Graphite, Spiral Wound

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-1472 - Human Engineering

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385620 - Handwheels for Valves

803-1385965 - 3-Way By-Pass Valve, 1/2-1, BW & SW 900 Lb ASME

803-2177525 - 1/4-2 BW & SW 900 Lb ASME Valve

803-5184193 - Steel Globe Valves, 1/4 to 2 nps, 1500 Class

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVSEA PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

S9086-RK-STM-010/505 - Naval Ships' Technical Manual Chapter 505, Piping Systems

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form)

ASME B1.5 - Acme Screw Threads

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ASME B16.20 - Metallic Gaskets for Pipe Flanges: Ring-Joint, Spiral-Wound, and Jacketed

ASME B16.34 - Valves Flanged, Threaded and Welding End

(Copies of these documents are available online at www.asme.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM A105/A105M - Standard Specification for Carbon Steel Forgings for Piping Applications

ASTM A106/A106M - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service

ASTM A182/A182M - Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service

ASTM A193/A193M - Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications

ASTM A194/A194M - Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both

ASTM A216/A216M - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service

ASTM A217/A217M - Standard Specification for Steel Castings, Martensitic Stainless and Alloy, for Pressure-Containing Parts, Suitable for High-Temperature Service

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A312/A312M - Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes

ASTM A335/A335M - Standard Specification for Seamless Ferritic Alloy-Steel Pipe for High-Temperature Service

ASTM A351/A351M - Standard Specification for Castings, Austenitic, for Pressure-Containing Parts

ASTM A387/A387M - Standard Specification for Pressure Vessel Plates, Alloy Steel, Chromium-Molybdenum

ASTM A515/A515M - Standard Specification for Pressure Vessel Plates, Carbon Steel, for Intermediate- and Higher-Temperature Service

ASTM A516/A516M - Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service

ASTM A564/A564M - Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless Steel Bars and Shapes

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ASTM B168	-	Standard Specification for Nickel-Chromium-Iron Alloys (UNS N06600, N06601, N06603, N06690, N06693, N06025, N06045, and N06696), Nickel-Chromium-Cobalt-Molybdenum Alloy (UNS N06617), and Nickel-Iron-Chromium-Tungsten Alloy (UNS N06674) Plate, Sheet, and Strip
ASTM B700	-	Standard Specification for Electrodeposited Coatings of Silver for Engineering Use
ASTM F718	-	Standard Specification for Shipbuilders and Marine Paints and Coatings Product/Procedure Data Sheet
ASTM F2168	-	Standard Specification for Packing Material, Graphitic, Corrugated Ribbon or Textured Tape, and Die-Formed Ring

(Copies of these documents are available online at www.astm.org.)

MANUFACTURERS STANDARDIZATION SOCIETY (MSS) OF THE VALVE AND FITTINGS INDUSTRY

MSS SP-25	-	Standard Marking System for Valves, Fittings, Flanges, and Unions
MSS SP-45	-	Bypass and Drain Connections
MSS SP-61	-	Pressure Testing of Valves
MSS SP-91	-	Guidelines for Manual Operation of Valves

(Copies of these documents are available online at http://msshq.org/Store/index.cfm.)

SAE INTERNATIONAL

SAE AS15003	-	Fitting, Lubrication, Hydraulic, Surface Check, ⅛ Pipe Threads, Steel, Type III
SAE AS35411	-	Fittings, Lubrication
SAE J1086	-	Numbering Metals and Alloys

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Standard and special rated valves furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.4).

3.2 Materials. Materials shall be as specified in table I. Materials for parts other than those listed in table I shall be selected to prevent galling, seizing, or excessive wear of operating parts, and require approval of the cognizant activity or agency.

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TABLE I. List of materials. [1/]

Name of part	Material form	Applicable documents			
		Composition A	Composition B	Composition D [2/]	Composition E
Body, bonnet, and yoke	Forgings or castings	ASTM A182/A182M, grade F 22 (UNS K21590) ASTM A217/A217M, grade WC 9 (UNS J21890)	ASTM A182/A182M, grade F 11 (UNS K11572) ASTM A217/A217M, grade WC 6 (UNS J12072)	ASTM A105/A105M, (UNS K03504) ASTM A216/A216M, grade WC B (UNS J03002)	ASTM A182/A182M, grade F 304 (UNS S30400) or grade F 216 (UNS S31600) ASTM A351/A351M, grade CF3 (UNS J92500) or grade CF3M (UNS J92800)
Seat ring	Forgings, castings, or plate	ASTM A182/A182M, grade F 22 (UNS K21590) ASTM A217/A217M, grade WC 9 (UNS J21890) ASTM A387/A387M, grade 22, class 2 (UNS K21590)	ASTM A182/A182M, grade F 11 (UNS K11572) ASTM A217/A217M grade WC 6 (UNS J12072) ASTM A387/A387M, grade 11, class 2 (UNS K11789)	ASTM A105/A105M (UNS K03504) ASTM A216/A216M, grade WC B (UNS J03002) ASTM A515/A515M, grade 70 (UNS K02700) ASTM A516/A516M, grade 70 (UNS K02700)	ASTM A182/A182M, grade F 304L (UNS S30403) or grade F 316L (UNS S31600) ASTM A351/A351M, grade CF3 (UNS J92500) or grade CF3M (UNS J82800) ASTM A240/A240M, grade 304 (UNS S30400) or grade 316 (UNS S31600)

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TABLE I. List of materials [1/] – Continued.

Name of part	Material form	Applicable documents			
		Composition A	Composition B	Composition D [2/]	Composition E
Retaining ring	Forgings, castings, or plate	ASTM A182/A182M, grade F 22 (UNS K21590) or grade F 11 (UNS K11572) ASTM A193/A193M, A105/A105M (UNS K03504) ASTM A217/A217M, grade WC 9 (UNS J21890) or grade WC 6 (UNS J12072) ASTM A216/A216M, grade WC B (UNS J03002) ASTM A182/A182M, grade F 304 (UNS S30400) or grade F 316 (UNS S31600) ASTM A515/A515M, grade 70 (UNS K03101) ASTM A387/A387M, grade 11, class 2 (UNS K11789) or grade 22, class 22 (UNS K21590)			ASTM A182/182M, grade F 304L (UNS S30403) or grade F 316L (UNS S31600) ASTM A351/A351M, grade CF3 (UNS J92500) or grade CFM3 (UNS J92800) ASTM A240/A240M, grade 304 (UNS S30400) or grade 316 (UNS S31600)
Stem	Barstock alloy steel	Cr 13 (see table VII)	Cr 13 (see table VII)	Cr 13 (see table VII)	ASTM A564/A564M, Type 630 (UNS S17400)
Bonnet fasteners bolts and studs	Barstock alloy steel	ASTM A193/A193M, grade B 16 (UNS K14072)		ASTM A193/A193M, grade B 7 (UNS H41400) or grade B 16 (UNS K14072)	ASTM A193/A193M, grade B 7 (UNS K14072)
Bonnet nuts	---	ASTM A194/A194M, grade 4 or 7 (UNS K14510)		ASTM A194/A194M, grade 2 H (UNS K04002) or grade 4 (UNS K14510)	ASTM A194/A194M, grade B 7 (UNS K14072)
Standard pressure seal rings and standard oversized pressure seal rings	Low carbon steel, silver plated	As specified herein (see 6.2).			
Alternate compression pressure seal rings [3/]	Graphite material	MIL-P-24503 or ASTM F2168			

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TABLE I. List of materials [1/] – Continued.

Name of part	Material form	Applicable documents			
		Composition A	Composition B	Composition D [2/]	Composition E
Gaskets (for flanged bonnets)	Spiral wound	MIL-G-24716, type I			
	Compressed Sheet	MIL-DTL-24696			
Packing	Graphitic	MIL-P-24503, MIL-P-24583			
Nuts and bolts (non-bonnet)	IAW MIL-DTL-1222	As specified herein (see 6.2), or materials in accordance with MIL-STD-1222.			
Valve trim	(see table VII)				

NOTES:

1/ Table I contains the commercial and military specifications and standards, and in addition, contains the corresponding SAE J1086 Unified Numbering System (UNS) designation. Reference to a specific UNS designation will permit the use of any one of the many alloy specifications and standards listed in SAE J1086 for that UNS number.

2/ The bonnet, yoke, and seat ring for Composition D valves may be in accordance with ASTM A217/A217M, grade WC 6 or ASTM A182/A182M, grade F 11.

3/ Graphite seal rings may be used in replacement of standard steel ones, with capped upper and lower seals to preclude packing extrusion.

3.2.1 Prohibited materials. Valves shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Construction. Unless otherwise specified herein, valves, valve parts, and construction features and parameters shall be in accordance with ASME B16.34.

3.4.1 Pressure temperature ratings. The design and pressure temperature rating shall be in accordance with ASME B16.34, except the maximum allowable temperature for composition D shall be 775 °F. The pressure seal valve shall be constructed in such a manner as to permit repair by the use of oversize rings.

3.4.2 End connections. End connections shall be in accordance with ASME B16.34; flanged, socket welded, or butt welded for the standard valves; and butt welded only for the special valves. Butt welding ends shall be compatible with the appropriate pipe schedule end preparation (see 6.2).

3.4.3 Flanges. Flanged end valves shall have flanges that are compatible with the corresponding mating flange for face preparation, number, and size of bolt holes and pressure rating (see 6.2).

3.4.4 Drains and bypasses. When specified (see 6.2), a drain or bypass shall be furnished and located as specified in 3.4.4.1 through 3.4.4.5 and ASME B16.34 and MSS SP-45. A drain shall consist of a nipple and drain valve. A bypass shall consist of connecting lines and a bypass valve.

3.4.4.1 Size. Standard drain and bypass sizes shall be as shown in table II. Nonstandard sizes shall be as specified (see 6.2).

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TABLE II. Standard drain and bypass sizes.

Valve sizes (inches)	Size of bypass (inch)	Size of bonnet drain (inch)
2½, 3, and 4	½	¼
5, 6, and 8	¾	¼
10, 12, 14, and 16	1	¼

3.4.4.2 Location. The location of drains and bypasses shall be specified (see 6.2) by referencing the letter designation in ASME B16.34 of the desired bosses, or, if nonstandard, in detail. Bosses shall be sufficient distance away from the seating area to allow welding of the replacement bypass line without damage to the valve seat. Threaded boss connections shall not be used.

3.4.4.3 Root connections. Drain and bypass line connections shall be welded to the boss in the valve body in accordance with S9074-AR-GIB-010/278. Materials for these lines shall be as follows:

a. Composition A – ASTM A335/A335M, grade P22.

b. Composition B – ASTM A335/A335M, grade P11.

c. Composition C – ASTM A106/A106M, grade B.

d. Composition E – ASTM A312/A312M, grade 304 (UNS 30400) or grade 316 (UNS 31600).

3.4.4.4 Bosses. Bosses shall be cast or forged integral with the body as specified (see 6.2). Construction and dimensions of bosses shall be in accordance with ASME B16.34, undrilled. Forged valve bodies with wall thickness in accordance with ASME B16.34 may have bosses built up with weld filler material selected in accordance with S9074-AR-GIB-010/278. Such bosses shall be liquid penetrant inspected and stress relieved in accordance with S9074-AR-GIB-010/278.

3.4.4.5 Valves. Drain and bypass valves shall be in accordance with 803-5184193, 803-2177525, or 803-1385965.

3.4.5 Water way diameters. Water way diameters of special valves shall be in accordance with table III. Water way diameters of standard classes and sizes shall be in accordance with the inside diameters of ASME B16.34.

TABLE III. Water way minimum diameters.

Valve sizes (inches)	Class 600 (inches)	Class 900 (inches)	Class 1,500 (inches)	Class 2,500 (inches)
2½	2⅛	1 5/16	1 5/16	1½
3	2⅝	2⅝	2⅜	1¾
4	3 7/16	3 7/16	3⅛	2½
5	4⅜	4 5/16	4 5/16	3
6	6	5	4¾	3¾
8	7⅞	6 9/16	6¼	5
10	9¾	8¼	7¾	6
12	10¼	10¼	9¼	7¼
14	11¼	11¼	10½	8
16	12⅞	12⅞	11¾	9

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3.4.6 Weights and center of gravity. Hand wheel actuated, welded, and type valves shall not exceed the maximum weights listed in tables IV and V. Weights listed are based on valves with welded ends and do not include weights of drains, bypasses, actuators, or other accessory items. Valves in classes and sizes not listed in tables IV and V do not have weight requirements. The valve weights in table IV are for class 600, 900, 1,500, and 2,500 cast gate valves, welded ends. The valve weights in table V are for the same valve design in the forged configuration. Valves with weights of 100 pounds or greater shall be marked with the vertical, longitudinal, and transverse location of the center of gravity, as specified (see 6.2).

TABLE IV. Maximum weights for cast body valves.

Valve size (inches)	Class 600 (pounds)	Class 900 (pounds)	Class 1,500 (pounds)	Class 2,500 (pounds)
2½	45	80	80	100
3	70	105	120	170
4	110	165	225	335
5	180	260	350	610
6	285	425	560	840
8	615	810	1,025	1,440
10	915	1,130	1,880	2,490
12	1,405	1,750	2,650	3,850
14	1,900	2,360	3,050	4,410
16	2,575	2,900	4,200	6,500

TABLE V. Maximum weights for forged body valves.

Valve size (inches)	Class 600 (pounds)	Class 900 (pounds)	Class 1,500 (pounds)	Class 2,500 (pounds)
2½	58	104	104	130
3	91	136	156	221
4	143	214	292	435
5	234	338	455	793
6	370	552	728	1,092
8	799	1,053	1,332	1,872
10	1,189	1,469	2,444	3,237
12	1,826	2,275	3,445	5,005
14	2,470	3,068	3,965	5,733
16	5,362	3,770	5,460	8,450

3.4.7 Flow coefficient. The flow coefficient (CV) for the 600 to 2,500 rated valves shall be as specified in table VI. The flow rates are based on values for valves in the fully open position.

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TABLE VI. Minimum CV.

Valve size (inches)	Class 600 (pounds)	Class 900 (pounds)	Class 1,500 (pounds)	Class 2,500 (pounds)
2½	355	295	295	141
3	542	542	444	189
4	958	958	791	409
5	1,599	1,554	1,544	558
6	3,107	2,157	1,947	915
8	5,540	3,847	3,489	1,672
10	8,493	6,080	5,366	2,301
12	9,740	9,740	7,933	3,406
14	11,734	11,734	10,221	4,593
16	15,369	15,369	12,800	5,074

3.4.8 Shock resistance. Valves and valves with attached gear or power actuators shall withstand high impact shock in accordance with grade A, hull-bulkhead mounted principal unit, complete assembly, class I, type C of MIL-DTL-901. Valves with gear or power actuators shall withstand these requirements with such actuators attached.

3.4.9 Vibration. Valves shall withstand the vibration requirements of MIL-STD-167-1. Valves with gear or power actuators shall withstand the vibration requirements of MIL-STD-167-1 with such actuators attached.

3.4.10 Bonnet.

3.4.10.1 Flanged, pressure seal, and seal-welded membrane pressure boundary bonnets. Class 150, 300, and 400 valves shall have flanged bonnets. Class 600, 900, 1,500, and 2,500 valves shall have pressure seal or seal-welded membrane pressure boundary bonnets.

3.4.10.2 Flanged bonnets. Flanged bonnets shall be of the male and female type, or small tongue and groove type, with spiral wound gaskets in accordance with MIL-G-24716.

3.4.10.3 Pressure seal bonnets. Seal rings for pressure seal bonnets shall provide a seal either by plastic or elastic expansion. The bonnet seal ring region of the valve body shall be inlaid with material in accordance with ASTM B168.

3.4.10.3.1 Plastic expansion seal rings. Seal rings sealing by plastic expansion shall have a Brinell hardness number (Bhn) of 100 maximum and be silver plated in accordance with ASTM B700, grade A.

3.4.10.3.2 Elastic expansion seal rings. Seal rings sealing by elastic expansion shall have a Bhn of 140 maximum and be silver plated in accordance with ASTM B700, grade A.

3.4.10.3.3 Graphitic material seal rings. Graphite seal rings may be used in replacement of standard steel ones, with capped upper and lower seals to preclude packing extrusion.

3.4.10.4 Seal-welded body to bonnet pressure boundary designs. Valves that employ seal-welded pressure boundary construction between the body and bonnet shall be constructed such that the seal weld is isolated from the structure and mechanical loading and only subjected to temperature and pressure transients. Valves employing a seal-welded external boundary shall reference a S9074-AR-GIB-010/278 approved welding procedure on the nameplate.

3.4.11 Yoke. Valves shall be of the outside screw and yoke construction (see 6.3.1).

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3.4.12 Stuffing boxes and glands.

3.4.12.1 Stuffing boxes. Stuffing boxes shall be of a depth sufficient to accommodate packing that will assure a pressure tight seal at the rated pressure and temperature of the valve and to assure a positive engagement between the stuffing box and gland.

3.4.12.2 Glands. Stuffing box glands shall be secured and adjusted by a threaded arrangement which, by adjustment, will vary the compression of the stem packing to seal the stem penetration under all operating conditions. Bearing surface between the gland follower and the gland flange shall be spherically finished.

3.4.13 Valve trim.

3.4.13.1 Stem. Valve stems shall have Acme-type threads in accordance with ASME B1.5 for valve operation and Unified National Standard threads for hand wheel attachments in accordance with ASME B1.1. Unless otherwise specified (see 6.2), the valve shall have a rising stem construction.

3.4.13.2 Disc/stem. Discs shall be of a double face construction. Discs in valves with ratings up to and including class 400 shall be of either the solid or flexible construction. Discs in valves of classes 600 to 2,500 shall be of the flexible construction. Valve construction shall include disc guide slots and the valve body shall have disc guides for the full length of the disc travel.

3.4.14 Main seat. Seat rings for classes 150 to 400 may be rolled in, threaded, or welded in place. Seat rings for classes 600 to 1,500, and for all classes for use in steam systems, shall be seal-welded if threaded or welded in place. When seat rings are threaded in valves intended for steam service, the seat ring shall be seal-welded circumferentially so as to prevent leakage past the seat rings.

3.4.14.1 Back seat. Valves shall have a positive back seat between the stem and bonnet only. Valves of class 600 and higher shall have a hard facing (HF) back seat that shall be the same material as the main seat in table VII (see 3.4.14.3). (Note that the term “hard facing” is sometimes referred to as “hard surfacing” in other welding publications, including S9074-AR-GIB-010/278.)

3.4.14.2 Valve trim material. Unless otherwise specified (see 6.2), valve trim material shall be in accordance with tables I and VII.

TABLE VII. Valve trim materials.

Service	Valve trim symbols 1/		Temperature limitations (°F)
	Disc 2/	Seat 2/	
Steam 3/	HF 3/	HF	1,050
Water 4/	Ni-Cu HF	Cr 13 HF	750 1,050
Oil	Ni-Cu Cr 13 HF	Cr 13 Cr 13 HF	500 1,000 1,050

NOTES:

1/ HF may be substituted for other seats and discs under all temperature and service conditions.

2/ Where unlike seats and discs are used, either the seat or the disc may be Cr 13 or Ni-Cu at the manufacturer’s option.

3/ Steam valves shall have HF seats and discs.

4/ Class 600 and higher water valves shall have HF seats and discs.

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3.4.14.3 Hard facing (HF). Welding materials shall be in accordance with type MIL-RCoCr-A-1, -2, and -3 of MIL-R-17131. The minimum Rockwell C 38 finished thickness of HF seating surfaces shall be 3/32 inch.

3.4.15 Valve hand wheels.

3.4.15.1 Rotation. Valves shall close by clockwise rotation of the hand wheel when facing the hand wheel.

3.4.15.2 Material and construction. Hand wheels 11 inches in diameter and smaller shall be manufactured from steel, ductile iron, malleable iron, or aluminum. Hand wheels 12 inches in diameter and larger shall be aluminum alloy or cast steel in accordance with the general dimensions of 803-1385620. Steam hand wheel connection shall be in accordance with 803-1385620.

3.4.15.3 Size. Hand wheels of valves and valve actuators shall be sized so that a tangential force of not more than the value specified in table VIII is required to be exerted on the rim of the hand wheel to effect tight closure (see 4.6.6.2). Valve operating devices or actuators may be necessary to accomplish effective closure or to allow compliance with human factor torque limitations in accordance with MIL-STD-1472. Such actuators shall be in accordance with MSS SP-91. Note that mounting orientation affects human factors.

TABLE VIII. Maximum allowable tangential force to seat valves based on valve hand wheel size.

Hand-wheel diameter (inches)	Maximum allowable tangential force to seat valve (pounds)	Maximum allowable torque to seat valve (foot-pounds)	Maximum allowable tangential force to open valve (pounds)	Maximum allowable torque to open valve (foot-pounds)
2	90	8	40	3
3	98	12	40	5
4	106	18	40	7
5	112	23	40	8
6	118	30	40	10
7	121	35	40	12
8	124	41	40	13
9	127	48	40	15
10	130	54	40	17
11	133	61	40	18
12	135	68	40	20
14	138	81	40	23
16	141	94	40	27
18	144	110	40	30
21	147	131	40	35
24	150	150	40	40
27	150	169	40	45
30	150	188	40	50
36	150	225	40	60

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3.4.16 Gear actuators. When specified (see 6.2), manual gear actuators shall be provided in accordance with MSS SP-91. The actuators shall be ductile iron, nodular iron, or steel.

3.4.16.1 Power actuators. When specified (see 6.2), power actuators shall be provided. Valves shall be constructed to assure means of mounting power actuators on the valves. When valves are provided with electric actuators, they shall be in accordance with DOD-V-24657.

3.4.17 Seat leakage. Unless otherwise specified (see 6.2), allowable seat leakage is considered to be leakage of water not in excess of ⅓ liquid ounce per hour per inch of nominal pipe size (nps) (see 4.6.6.2).

3.4.18 Lubrication. Yoke bushings shall be equipped with a ⅛-inch threaded or push design, type III lubricating fitting in accordance with SAE AS35411 and SAE AS15003.

3.4.19 Welding and fabrication.

3.4.19.1 Welding. Welding shall be in accordance with S9074-AR-GIB-010/278.

3.4.19.2 Fabrication. Fabricated assemblies shall be stress relieved as units prior to finish machining. Stress relieving shall be done in accordance with S9074-AR-GIB-010/278.

3.4.20 Maintainability. Valve construction shall assure that assembly and disassembly of the valve can be accomplished without the need for special training or special tools. Configurations shall be such as to permit the use of a portable boring machine to repair the bonnet inlay of the valve with the valve in place in the piping system.

3.4.21 Interchangeability. Parts shall not be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.4.22 Corrosion protection. When specified (see 6.2), valve bodies shall be metal sprayed for corrosion protection in accordance with ASTM F718.

3.5 Body markings and identification plates.

3.5.1 Body markings. Valve bodies shall have the composition, class, and the manufacturer's name or trademark cast or forged integral with the valve body. When necessary, metal stamping will be permitted on the neck of the valve body or other similar surface not subjected to high stresses in service. In addition to the requirements herein, markings shall be in accordance with MSS SP-25.

3.5.2 Identification plates. Identification plates in accordance with MIL-DTL-15024 and MIL-P-15024/5, made of corrosion-resisting steel or brass, shall be permanently fastened to a part of the valve not subjected to working pressure, preferably the yoke. Identification plates shall include or provide space for the following data:

a. Manufacturer's name or trademark.
b. Size of valve and class.
c. Body and bonnet material composition.
d. Valve trim identification (stem-disc-seat).
e. Manufacturer's identification number (optional).
f. Manufacturer's drawing number.
g. MIL-DTL-18110.
h. Component identification number (CID), if known.
i. National stock number (NSN), if known.
j. Membrane seal welding and repair procedure in accordance with 3.4.10.4.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows (see table IX):

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

TABLE IX. Examination and tests.

Characteristics	Requirement	Qualification	Conformance
Materials	3.2	4.5.1	---
Construction	3.4	4.5.2, 4.6.1	4.5.2
Pressure temperature ratings	3.4.1	4.6.2, 4.6.3, 4.7	4.6.2
End connections	3.4.2	4.5.2	---
Flanges	3.4.3	4.5.2	---
Drains and bypasses	3.4.4	4.5.2	---
Water way diameters	3.4.5	4.5.2	---
Weight and center of gravity	3.4.6	4.5.2	4.5.2
CV	3.4.7	4.5.2	---
Shock resistance	3.4.8	4.6.4	---
Vibration	3.4.9	4.6.5	---
Bonnet	3.4.10	4.5.2	---
Yoke	3.4.11	4.5.2	---
Stuffing box and gland	3.4.12	4.6.6	---
Trim	3.4.13	4.6.6	---
Main seat	3.4.14	4.6.6	---
Hand wheels	3.4.15	4.5.2	---
Gear actuators	3.4.16	4.6.4	---
Seat leakage	3.4.17	4.6.6	---
Lubrication	3.4.18	4.5.2	---
Welding and fabrication	3.4.19	4.5.2	---
Maintainability	3.4.20	4.7	---
Interchangeability	3.4.21	4.7	---
Corrosion protection	3.4.22	4.5.2	---
Body markings and identification plates	3.5	4.5.2	4.5.2

4.2 Qualification inspection. Qualification inspection of the special and standard valves shall be conducted at a laboratory acceptable to NAVSEA. Qualification inspection (see 6.4) shall consist of all the examinations and tests specified in table IX.

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4.2.1 Inspection samples. Sample valves submitted for qualification inspections shall be of the size, class, and composition specified in table X. Valve sizes that have passed the qualification examinations and tests as specified in table IX will qualify or certify other valve sizes of identical class and composition as listed in table X provided that variations in construction throughout the various sizes and classes are demonstrated by analyses to be approved by NAVSEA.

TABLE X. Sample valve sizes, classes, and composition for qualification inspection.

Size (inches)	Class and composition	Certifies these valve sizes, classes, and compositions		
		Composition	Classes	Size (inches)
4	1,500, B	A, B, and D	600, 900, 1,500, and 2,500	2½ to 4 inclusive
	400, D	D	150, 300, and 400	
	400, E	E		
8	1,500, B	A, B, and D	600, 900, 1,500, and 2,500	5 to 8 inclusive
	400, D	D	150, 300, and 400	
	400, E	E		
10	1,500, B	A, B, and D	600, 900, 1,500, and 2,500	9 and 10 inclusive
	400, D	D	150, 300, and 400	
	400, E	E		
All sizes above 10	1,500, B	A, B, and D	600, 900, 1,500, and 2,500	12, 14, and 16
	400, D	D	150, 300, and 400	
	400, E	E		

NOTE:
1. Testing of lower pressure classes will certify only lower pressure class valves.

4.3 Conformance inspection. Conformance inspection shall consist of the examinations and tests specified in table IX. Each special and standard sample valve shall be examined.

4.4 Sampling for examinations and tests.

4.4.1 Lot. A lot shall consist of valves of the same size, class, and composition offered for delivery at the same time.

4.4.2 Sample. A random sample of valves shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and Acceptance Quality Limit (AQL), as specified (see 6.2) for the visual and dimensional examination specified (see 4.5).

4.5 Examinations.

4.5.1 Examination of materials. Materials used in the manufacture of valves shall be examined to determine conformance to 3.2 and table I, as applicable.

4.5.2 Visual and dimensional examination. Each special and standard sample valve (see 4.4) shall be examined to determine compliance with all requirements of this specification not involving tests. Any sample valve that fails to meet these requirements shall be rejected.

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4.6 Test methods.

4.6.1 Nondestructive testing. Nondestructive testing shall consist of the following radiographic, magnetic particle, and liquid penetrant inspections.

4.6.1.1 Radiographic. Pressure-containing castings of valves for class 600 and above shall be radiographically inspected in accordance with the following:

a. Radiographic inspection shall conform to T9074-AS-GIB-010/271.

b. Radiographic acceptance shall conform to S9074-AR-GIB-010/278.

c. Radiographic coverage shall be 100 percent of total area.

4.6.1.2 Magnetic particle. Pressure-containing forgings of valves in class 600 and above shall be magnetic particle inspected in accordance with T9074-AS-GIB-010/271. Forgings shall be free from flaws such as seams, cracks, laps, porosity, scale, flakes, and all other defects detrimental to the suitability of the forging for the service intended.

4.6.1.3 Liquid penetrant. Hard faced seating surfaces shall be liquid-penetrant inspected in accordance with T9074-AS-GIB-010/271 after rough machining and shall be free of cracks or crack-like defects.

4.6.2 Cold tests. Cold tests shall be performed as follows:

a. Operation – 2,000 cycles.

b. Hydrostatic – shell and closure tests (see 4.6.6).

4.6.2.1 Cold operational testing. Valves shall be cycled a minimum of 2,000 times in a cold loop, pressurized to not less than 100 percent of rated pressure at 100 °F in accordance with ASME B16.34 and 1.2.2. There shall be a flow of water through the valve when it is open during the test, with the quantity dependent upon the test facility. During the test of the valves, there shall be a differential pressure of not less than 100 percent of rated pressure at 100 °F, in accordance with ASME B16.34 and 1.2.2, existing across the valve at the beginning of each opening. The torque required to open the valve shall be determined at the start, after 1,000 cycles, and after 2,000 cycles. A seat leakage test shall be performed in accordance with 4.6.6.2 at the start, after 1,000 cycles, and after 2,000 cycles. The leakage rate shall be in accordance with 3.4.17.

4.6.3 Hot tests. Hot tests shall be conducted with steam temperature and pressure in accordance with table XI and shall be performed in the following sequence:

a. Thermal shock.

b. Operation – 200 cycles.

4.6.3.1 Thermal shock tests. Valves shall be thermal shocked 10 times by reducing the steam temperature in table XI by a minimum of 35 percent in not more than 30 seconds. Steam temperatures shall be measured at a point as close to the valve as possible, preferably with the temperature sensing inside the valve. In addition to the steam temperature, a thermocouple shall be attached to the exterior of the valve in a zone of maximum temperature and minimum wall thickness. This thermocouple shall measure each thermal shock cycle to determine the temperature of the valve at the start of the cycle, the minimum temperature obtained during the cycle, the time required to reduce the valve to the minimum temperature, and the time required to restore the valve to the maximum temperature.

4.6.3.2 Operational tests. Steam temperature and pressure shall be passed through the valve for a period of no less than 48 hours. During this period, at approximately ⅓ intervals, the valve shall be closed and cooled to ambient temperature a total of three times. After cooling, the valve shall be opened and the torque required to open the valve shall be determined. Subsequently, the steam flow shall be resumed, the valve shall be cycled from open to closed a minimum of 200 times, and the operating torques shall be determined at each 100 cycles.

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TABLE XI. Steam temperature and pressure for hot test of valve.

Class	Pressure (lb/in^2) ±10%	Temperature (°F) +10%, -0%	Applicable valve composition
250	150	550	A, B, D, E
300	300	600	A, B, D, E
400	400	650	A, B, D, E
600	600	750	A, B, D
900	900	850	A, B, D
1,500	1,200	950	A, B
2,500	1,200	950	A, B

4.6.4 Shock test. Valves shall be subjected to the grade A shock test in accordance with MIL-DTL-901 to determine conformance to the requirements specified in 3.4.8. Valves shall be shock tested while hydrostatically pressurized to 100 percent of the 100 °F rating as specified in 1.2.2 throughout at the full open position and on the inlet side at the full closed position. Testing medium shall be clean tap water with no additive other than traces of cutting oil at a temperature between 40 and 150 °F. Momentary malfunction at the time of impact blow is permitted. Permanent deformation, misalignment, or functional impairments shall be cause for rejection. The valve shall be cycled 10 times with the allowable torque of table VIII and the seat leakage test shall be performed in accordance with 4.6.6.2 immediately after the completion of the shock tests. The leakage rate shall be in accordance with 3.4.17.

4.6.5 Vibration test. Valves shall be subjected to the type I vibration test in accordance with MIL-STD-167-1 to determine conformance to the requirements specified in 3.4.9.

4.6.6 Hydrostatic tests. Valves shall be subjected to the tests specified in 4.6.6.1 for strength and porosity and in 4.6.6.2 for tightness conducted in accordance with MSS SP-61. Water temperature shall not exceed 100 °F.

4.6.6.1 Shell test. Valves shall be given a shell test at a gauge pressure not less than 1½ times the 100 °F rating, rounded up to the next 25 lb/in^2 increment. The test shall be made with water, which may contain a corrosion inhibitor, or with kerosene, or with other suitable fluid, provided such fluid does not have a viscosity greater than water, at a temperature not above 125 °F. There shall be no visually detectable leakage through pressure boundary walls. Test duration shall be not less than as follows:

Valve size nps (inches)	Test time (seconds)
2½ to 8	60
10 and larger	180

Tests shall be conducted with the valve in the partially open position. Leakage through the stem packing, not under pressure, shall not be cause for rejection (S9086-RK-STM-010/505 may provide guidance).

4.6.6.2 Closure test. Following the shell test, valves shall be given a closure test. Each valve shall be given a closure test using fluid described in 4.6.6.1 at a pressure not less than 110 percent of the 100 °F pressure rating. The valve shall be main seated using a seating torque in accordance with hand wheel forces in table VIII. The test pressure shall be applied alternately to each side of the valve disc. Leakage rate of fluid test shall not exceed ⅓ liquid ounce per hour per inch of nps (see 3.4.17). The duration of the closure test shall be the same as specified in 4.6.6.1.

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4.6.6.3 Special conditions. Valves designed for operating conditions that have the pressure differential across the closure member limited to values less than the 100 percent pressure rating and have closure members or actuating devices (direct, mechanical, fluid, or electrical) that would be subject to damage at high differential pressures, shall be tested as described above. However, the closure test may be reduced to 110 percent of the maximum specified closed position differential pressure.

4.7 Examination after tests. Valves shall be disassembled and visually and dimensionally examined for damage and wear. Any valve or sample lot meeting the defect criteria of table XII shall be rejected. Disassembly and reassembly shall be performed to determine the practicability of maintaining a valve of this design for shipboard use (see 3.4.20). The maintainability demonstration shall be conducted by removing and replacing the pressure seal ring, or seal-welded flexible seal ring, disc, and stem. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during qualification testing. The demonstration procedures in MIL-HDBK-470 may be used. Particular emphasis shall be directed toward the possibility of the loss of small parts as a result of maintenance.

TABLE XII. Classification of defects.

Categories	Critical Defects
001	Classifications not as specified (see 1.2).
003	Material not as specified (see 3.2).
004	Pressure temperature rating not as specified (see 3.4.1).
Major Defects	
102	End connections not as specified (see 3.4.2).
103	Flanges not as specified (see 3.4.3).
104	Drains and bypasses not as specified (see 3.4.4).
105	Water way diameters not as specified (see 3.4.5).
106	Weight and center of gravity not as specified (see 3.4.6).
107	CV not as specified (see 3.4.7).
108	Shock not as specified (see 3.4.8).
109	Vibration not as specified (see 3.4.9).
110	Bonnet not as specified (see 3.4.10).
111	Yoke not as specified (see 3.4.11).
112	Stuffing box and gland not as specified (see 3.4.12).
113	Trim not as specified (see 3.4.13).
114	Main and back seat not as specified (see 3.4.14).
115	Hand wheels not as specified (see 3.4.15).
116	Gear actuators not as specified (see 3.4.16).
117	Seat leakage not as specified (see 3.4.17).
118	Lubrication not as specified (see 3.4.18).
119	Welding and fabrication not as specified (see 3.4.19).
120	Maintainability not as specified (see 3.4.20).
121	Interchangeability not as specified (see 3.4.21).
122	Corrosion protection not as specified (see 3.4.22).

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4.8. Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.1 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Steel gate valves are intended for use in steam, water, and oil service for use in naval shipboard steam, water, and oil service. Existing stock of steel gate valves manufactured to MIL-DTL-18110F(SH) dated 19 February 1991 are acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity required.
c. Composition, rating and class, and size required (see 1.2).
d. Service conditions.
e. Drawings applicable for list of materials (see table I).
f. Requirement for prohibited material verification (see 3.2.1 and 4.8).
g. End connection preparation (see 3.4.2).
h. End flanges: face preparation, number, size of bolt holes, and pressure rating (see 3.4.3).
i. Location of drains and bypasses (see 3.4.4 and 3.4.4.2).
j. Nonstandard sizes of drains and bypasses (see 3.4.4.1).
k. Boss requirements (see 3.4.4.4).
l. Center of gravity location (see 3.4.6).
m. Stem requirement if other than specified (see 3.4.13.1).
n. Valve trim material if other than specified (see 3.4.14.2).
o. Requirement for manual gear actuators (see 3.4.16).
p. Requirement for power actuators (see 3.4.16.1).
q. Allowable seat leakage if other than specified (see 3.4.17).
r. Corrosion protection required (see 3.4.22).
s. AQL of 1.0 (see 4.4.2).
t. Packaging requirements (see 5.1).

6.3 Definitions

6.3.1 Rising stem/outside screw and yoke. Rising stem/outside screw and yoke construction retains stem threads outside the valve. Rising stem/outside screw and yoke construction is recommended where high temperatures, corrosives, and solids in the line might damage stem threads inside the valve. When the hand wheel is turned, the stem rises as the yoke bushing engages the stem threads. The external threads enable easy lubrication; however, care must be taken to protect the exposed stem threads from damage. An advantage of rising stem valves is the ability to determine valve position by observing the position of the stem.

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6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 18110 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.5 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition of any valve under this specification:

a. Engineering drawings.
b. Materials listing.
c. Installation drawings.
d. Technical manual.
e. Test reports.
f. Certifications.
g. Onboard repair parts list.

6.6 Subject term (key word) listing.

Bonnet
Bosses
Flow coefficient
Hard facing
Outside screw
Rising stem
Valve seat
Water way
Yoke

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4820-2018-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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